

Section 4: Worksheet

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Background Information

Today is not only about being a good coder, but also about being CREATIVE! You will learn how to create a plot using R. For that, you will need to know the following information to respond to the next questions:

Column: SEX Record Respondent's Sex: 1. Male 2. Female

Column: S2 What is your race? 2. Black or African American

Column: S3 Are you Hispanic or Latino? 1. Yes, Mexican 2. Yes, Puerto Rican 3. Yes, Cuban 4. Yes, other Spanish background

Column: 3 On the whole, do you think immigration (people coming from other countries to LIVE in the US) has been a good thing or bad thing to the US IN THE PAST? 1. Good thing 2. Bad thing 3. Some good/Some bad 9. Don't Know/Refused

Column: 4 On the whole, do you think immigration (people coming from other countries to LIVE in the US) has been a good thing or bad thing to the US TODAY?*

1. Good thing
2. Bad thing
3. Some good/Some bad
4. Don't Know/Refused

Column: 9 As far as you know, do MOST immigrants in this group who live in your city or town have higher family incomes than you do, lower incomes, or are their incomes about the same as yours?

1. Higher
2. Lower
3. About the same
4. Don't know/Refused

Column: 10A Have recent immigrants made the overall quality of life in your city or town better or worse, or haven't they had much effect either way?

Column: 10C Have recent immigrants made the local crime situation better or worse, or haven't they had much effect either way?

Column: 10E Have recent immigrants made the local food, music, arts, and culture better or worse, or haven't they had much effect either way?

Column: 10F Have recent immigrants made the local economy in general better or worse, or haven't they had much effect either way?

Note that: 1. Better 2. Worse 3. Not much effect 4. Don't know 5. Refused

Before you start, remember to:

```
# loading my data
survey_data <- read.csv("survey_attitudes_immigrants.csv")
# load the packages you will need
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(ggplot2)
```

Question 1 (Choose One)

Bar Plots

a. Using the information we have on Black/African American, create a bar plot with their average response to Question 3, which is “On the whole, do you think immigration (people coming from other countries to LIVE in the US) has been a good thing or bad thing to the US IN THE PAST?” Order your bars from highest to lowest.

```
# First, I define labels for my variable Q3, since the raw data only has numbers
# These labels will make the plot easier to read.
labels_q3 <- c("1" = "Good thing",
               "2" = "Bad thing",
               "3" = "Some good/Some bad")

# Now I start working with my dataset, which I changed to "survey_data"
# I changed it because of an error a lot of you got. The error came up because:
# "data" is also the name of a built-in R function (from the utils package)
survey_data %>%
  # I filter the data to only include Black/African American respondents (S2 == 2).
  # I also exclude anyone who answered 9 on Q3, since that code represents missing or "Don't know".
  filter(S2 == 2, Q3 != 9) %>%

  # Next, I count how many people selected each option for Q3.
  # This gives me the frequency of responses within this group.
  count(Q3) %>%

  # I start building my plot with ggplot (You can use the R Code Guide I gave you)
  ggplot(aes(
    # Here I put the Q3 categories on the x-axis, AND I also reorder them by their counts (n).
    # I use reorder(as.factor(Q3), -n), which means the categories will be ordered
    # from the highest count to the lowest count (descending order).
    x = reorder(as.factor(Q3), -n),
```

```

# I put the counts on the y-axis.
y = n,

# I color the bars by the response category.
fill = as.factor(Q3))) +

# I add bars to the plot. Since I already calculated the counts with count(),
# I set stat = "identity" so ggplot uses those values directly.
geom_bar(stat = "identity") +

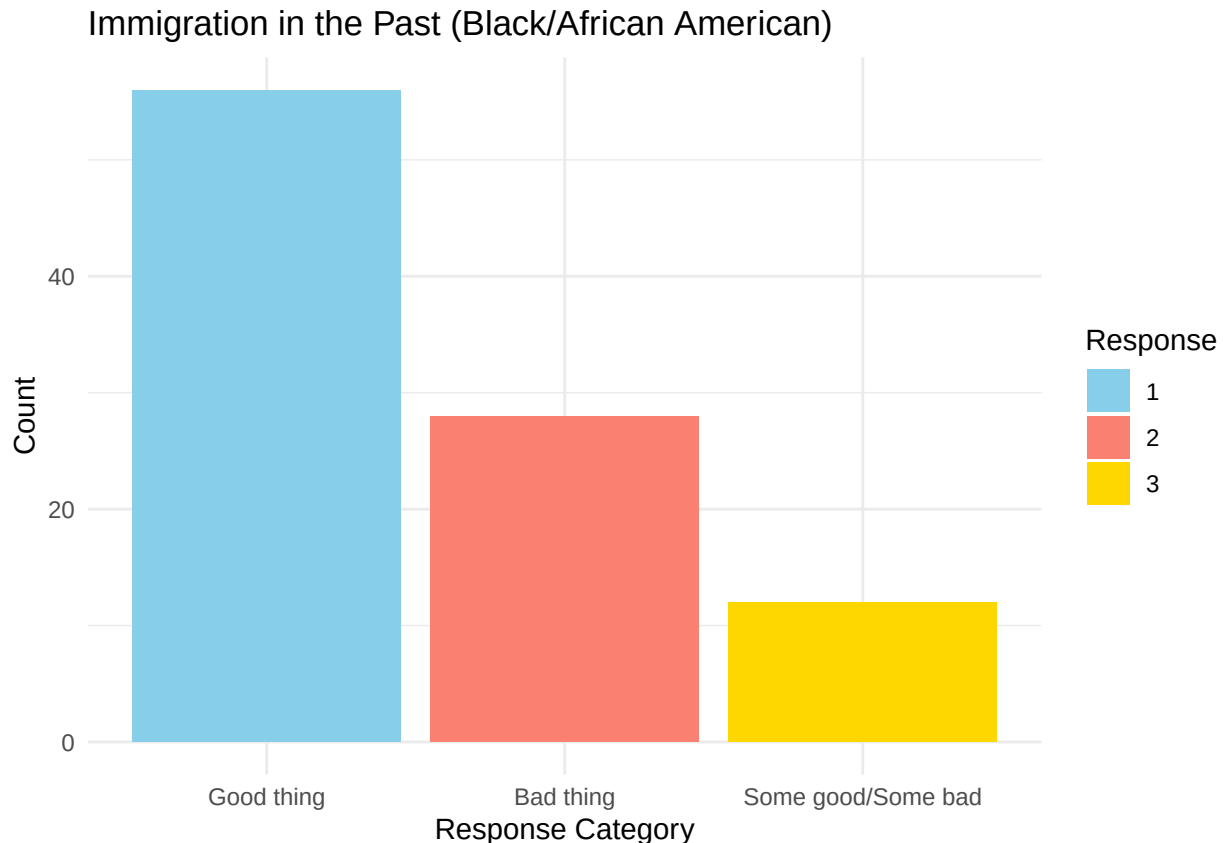
# I replace the numeric values on the x-axis with my custom labels ("Good thing", etc.).
scale_x_discrete(labels = labels_q3) +

# THE EXCITING PART: I choose my own colors for each response category to make the plot clearer.
scale_fill_manual(values = c("1" = "skyblue",
                             "2" = "salmon",
                             "3" = "gold")) +

# I add a title to describe what the plot shows,
# and I label the x-axis, y-axis, and legend.
labs(title = "Immigration in the Past (Black/African American)",
      x = "Response Category",
      y = "Count",
      fill = "Response") +

# Finally, I apply a minimal theme so the plot looks clean.
theme_minimal()

```



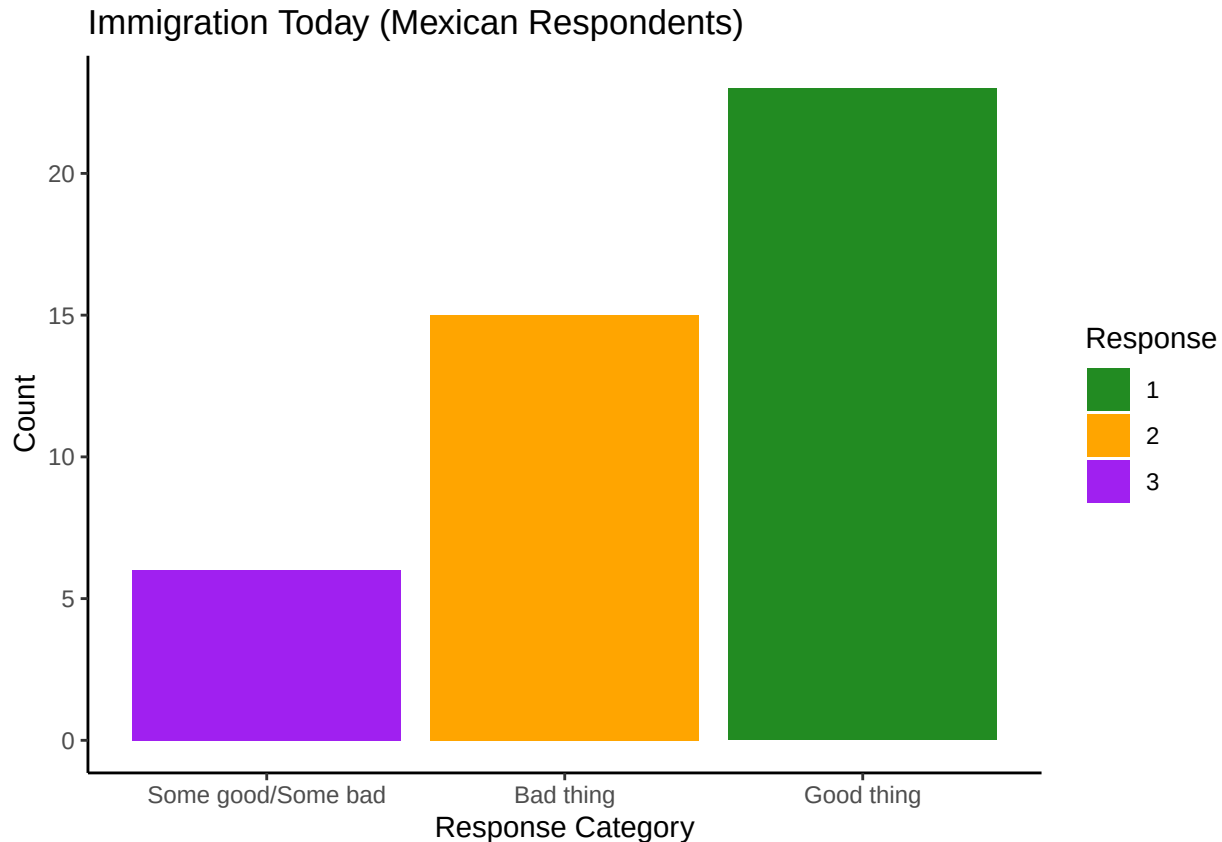
b. Using the information we have on Hispanic/Latino, pick one of the identities (Mexican, Puerto Rican, Cuban, Others), and create a bar plot with their average response to Question 4, which is “On the whole, do you think immigration (people coming from other countries to LIVE in the US) has been a good thing or bad thing to the TODAY?” Order your bars from lowest to highest.

```
# You do not have to start by defining the labels like this
# This is my personal preference for my code, but as long as you have define your # labels before you

# Define labels for Q4 so the categories show up with words instead of numbers.
labels_q4 <- c("1" = "Good thing",
               "2" = "Bad thing",
               "3" = "Some good/Some bad")

survey_data %>%
  filter(S3 == 1, Q4 != 9) %>%      # I keep only Mexican respondents and drop "Don't know"
  count(Q4) %>%                    # Count how many people chose each option
  ggplot(aes(x = reorder(as.factor(Q4), n), # I order the bars from lowest to highest
             y = n,
             fill = as.factor(Q4))) +
  geom_bar(stat = "identity") +     # Plot the counts directly
  scale_x_discrete(labels = labels_q4) +
  scale_fill_manual(values = c("1" = "forestgreen", # ASSIGN COLORS!
                               "2" = "orange",
                               "3" = "purple")) +
```

```
labs(title = "Immigration Today (Mexican Respondents)",
     x = "Response Category",
     y = "Count",
     fill = "Response") +
theme_classic()
```



Question 2 (Choose One)

Histogram

Create a histogram for:

a. Have recent immigrants made the local crime situation better or worse, or haven't they had much effect either way?

```
# Define labels for Q10C so the responses show up with words.
labels_10 <- c("1" = "Better",
               "2" = "Worse",
               "3" = "Not much effect")

survey_data %>%
  filter(Q10C < 4) %>% # I keep only valid responses (1-3)
  ggplot(aes(x = as.factor(Q10C), fill = as.factor(Q10C))) +
  geom_histogram(stat = "count", color = "black") + # Making bars showing how many chose each option
  scale_x_discrete(labels = labels_10) + # Replace numbers with labels
  scale_fill_manual(values = c("1" = "lightblue",
```

```

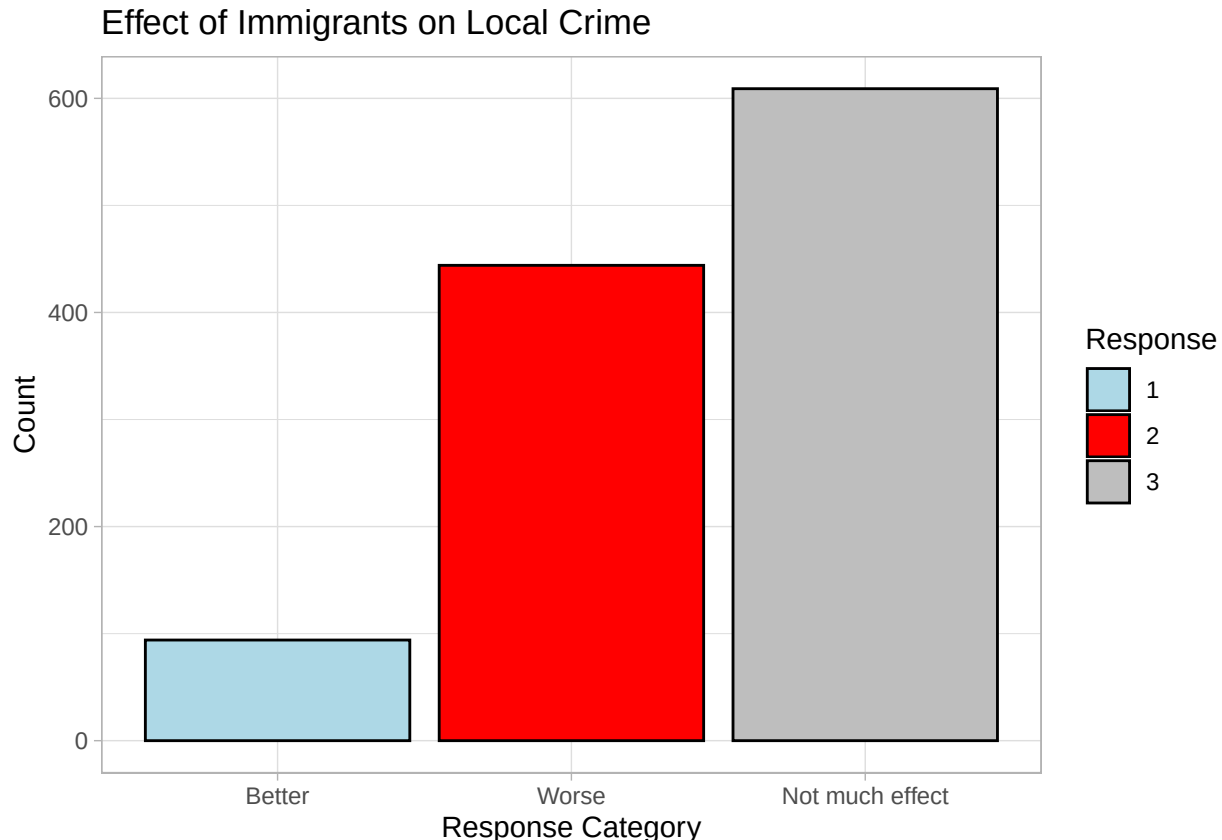
    "2" = "red",
    "3" = "gray")) +
labs(title = "Effect of Immigrants on Local Crime",
     x = "Response Category",
     y = "Count",
     fill = "Response") +
theme_light()

```

```

## Warning in geom_histogram(stat = "count", color = "black"): Ignoring unknown
## parameters: `binwidth`, `bins`, and `pad`

```



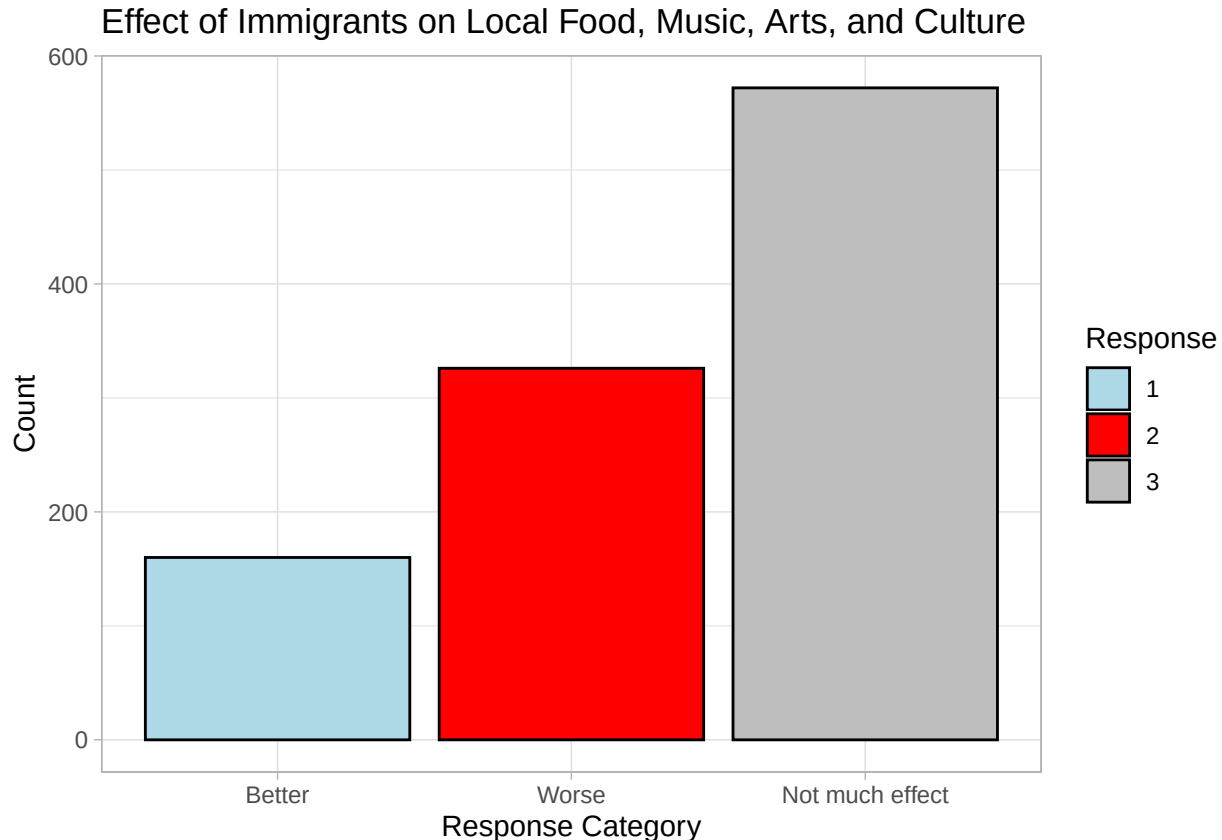
b. Have recent immigrants made the local food, music, arts, and culture better or worse, or haven't they had much effect either way?

```

# Start with your data!
survey_data %>%
  filter(Q10B < 4) %>% # I keep only valid responses (1-3)
  ggplot(aes(x = as.factor(Q10B), fill = as.factor(Q10B))) +
  geom_bar(color = "black") + # I plot counts directly for each response
  scale_x_discrete(labels = labels_10) +
  # Note that I reuse the labels I already created for Q10C
  # since Q10B uses the exact same response categories. We already created that
  # No need to create a new labels object when the categories are identical.
  scale_fill_manual(values = c("1" = "lightblue",
                                "2" = "red",
                                "3" = "gray")) +

```

```
labs(title = "Effect of Immigrants on Local Food, Music, Arts, and Culture",
     x = "Response Category",
     y = "Count",
     fill = "Response") +
theme_light()
```



Note that: 1. Better 2. Worse 3. Not much effect 4. Don't know 5. Refused

Question 3 (Pick one)

Density Plot

a. Using the information we have on Column 9 (“As far as you know, do MOST immigrants in this group who live in your city or town have higher family incomes than you do, lower incomes, or are their incomes about the same as yours?”), create a density plot that compares the distribution of responses between Male and Female respondents (Column D1).

```
# I define labels for Q9 (income perceptions) and for SEX (male/female).
labels_q9 <- c("1" = "Higher",
               "2" = "Lower",
               "3" = "About the same")

labels_sex <- c("1" = "Male",
               "2" = "Female")
```

```

survey_data %>%
  filter(Q9 < 9) %>%    # I keep only valid responses (exclude missing / DK)
  count(SEX, Q9) %>%    # Counting how many males and females gave each response
  group_by(SEX) %>%     # I group by sex so I can calculate percentages within each group
  mutate(prop = n / sum(n)) %>% # Here I convert counts to proportions (so each group adds up to 1)
  ggplot(aes(x = Q9, y = prop, fill = as.factor(SEX), color = as.factor(SEX))) +

  # Normally density plots are for continuous data, but here I use stat = "identity"
  # so I can show the categorical proportions in a density-style plot.

  geom_density(stat = "identity", alpha = 0.4) +

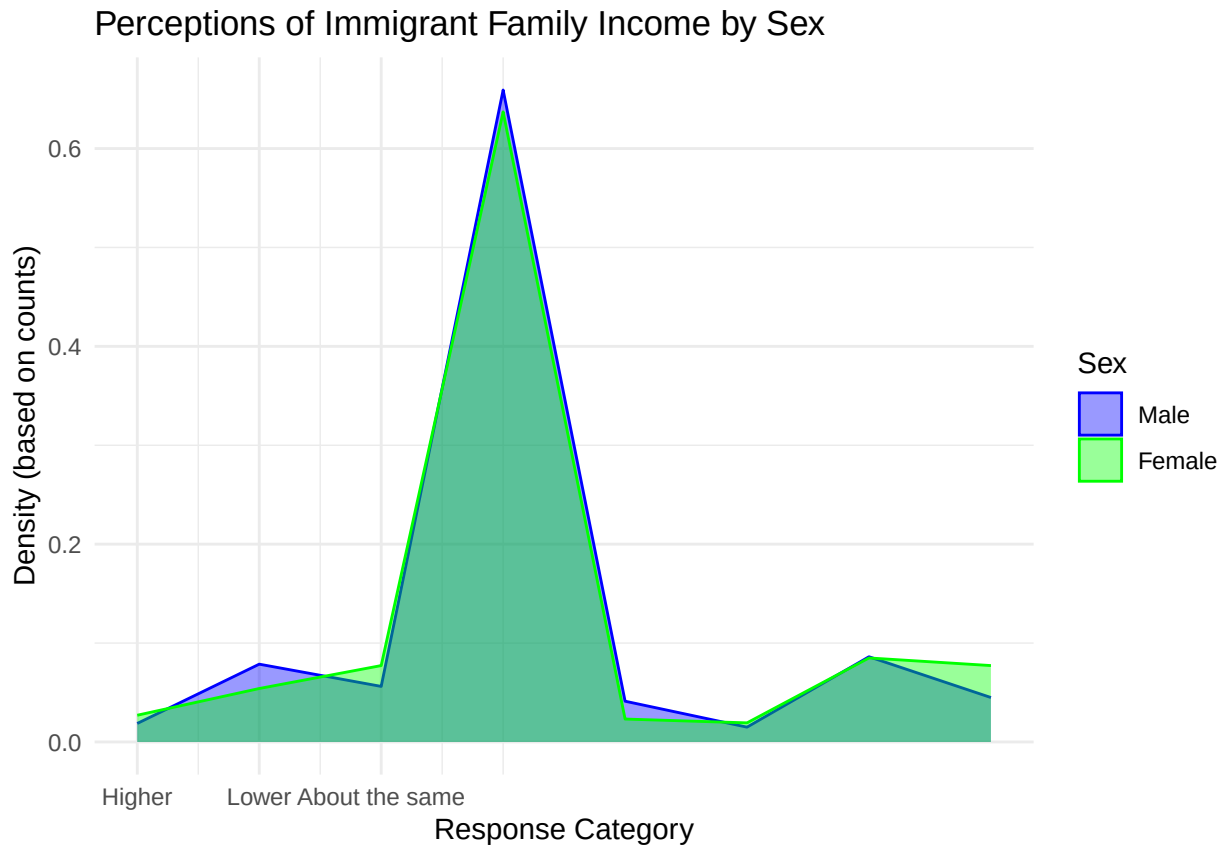
  # I relabel the x-axis ticks so the categories show as "Higher / Lower / About the same".
  scale_x_continuous(breaks = 1:3, labels = labels_q9) +

  # I add custom colors for Male and Female, and relabel the legend.
  scale_fill_manual(values = c("1" = "blue", "2" = "green"), labels = labels_sex) +
  scale_color_manual(values = c("1" = "blue", "2" = "green"), labels = labels_sex) +

  # I add a title, axis labels, and legend title.
  labs(title = "Perceptions of Immigrant Family Income by Sex",
        x = "Response Category",
        y = "Density (based on counts)",    # I clarify this is categorical density
        fill = "Sex",
        color = "Sex") +

  # I apply a minimal theme for a clean look.
  theme_minimal()

```

IN-

INTERPRETATION:

The density plot shows that both male and female respondents largely share the same perception about immigrant family incomes, with the vast majority believing they are about the same as their own. Only a small share think immigrant families have higher or lower incomes, and the distributions across sex overlap almost entirely. The only slight difference is that men are a bit more likely than women to say immigrants have lower incomes, but overall, perceptions are very similar between the two groups.

a. Using the information we have on Column 10F (“Have recent immigrants made the local economy in general better or worse?”), create a density plot that compares the distribution of responses between Male and Female respondents (Column D1).

```
survey_data %>%
  filter(Q10F < 4) %>% # I keep only valid responses (1-3)
  count(SEX, Q10F) %>% # I count how many male/female respondents gave each response
  group_by(SEX) %>% # I group by sex
  mutate(prop = n / sum(n)) %>% # I convert counts to proportions within each sex
  ggplot(aes(x = Q10F, y = prop, fill = as.factor(SEX), color = as.factor(SEX))) +
  geom_density(stat = "identity", alpha = 0.4) +
  scale_x_continuous(breaks = 1:3, labels = labels_10) +

  # I customize colors and legend labels for male/female
  scale_fill_manual(values = c("1" = "darkblue", "2" = "deeppink"), labels = labels_sex) +
  scale_color_manual(values = c("1" = "darkblue", "2" = "deeppink"), labels = labels_sex) +

  labs(title = "Effect of Immigrants on Local Economy by Sex",
```

```
x = "Response Category",
y = "Density (based on counts)", # I make clear this is a categorical density
fill = "Sex",
color = "Sex" +
```

```
theme_minimal()
```

